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Top Skills

Prompt Engineering
Agentic AI Development
Technical Project Leadership

Languages

English (Full Professional)
Spanish (Native or Bilingual)

Certifications

Graph Developer - Associate
Introduction to MongoDB
AWS Cloud Practitioner Essentials
Graph Developer - Professional
Object-Oriented Design

Honors-Awards

Academic Excellence

Patents

Systems and methods for graph management
Session-Based Data Storage for Chat-Based Communication Sessions
Systems and methods for bot dialog delegation

Maria Lucena

Director Of Architecture @ Fidelity Investments
Greater Philadelphia

Summary

I'm a Software Architect with over 15 years of experience leading distributed systems, cloud architectures, and high-scale API platforms. Throughout my career, I've grounded my work in solid engineering discipline, whether I am driving enterprise GraphQL Federation standards, designing REST APIs, or co-authoring patents for API gateway solutions. My approach to the evolving development landscape is highly practical. At the enterprise level, I focus on leveraging prompt design and agentic workflows to accelerate system delivery, recently using these methods to streamline a large-scale API inventory system. To keep my skills relevant, I stay deeply hands-on through independent engineering projects. In these personal endeavors, I architect solutions using Retrieval-Augmented Generation (RAG), Model Context Protocol (MCP), and custom AI skills to safely connect models with real-world data.

Experience

Fidelity Investments

11 years 2 months

Director of Engineering

March 2025 - Present (1 year 4 months)

New Jersey, United States

- Delivered a GraphQL enterprise solution integrating ~30 subgraphs. Drove firm-wide adoption by establishing architectural standards, leading learning sessions, and co-authoring a patent for GraphQL innovations.
- Architected a large-scale enterprise API inventory system, pragmatically applying AI-assisted development and agentic workflows to accelerate the engineering lifecycle and augment developer capabilities. The platform reached delivery in mid-2026.
- Driving enterprise data governance for the API ecosystem, defining and implementing robust standards based on industry-accepted frameworks.

Director Of Architecture

November 2021 - Present (4 years 8 months)

Hybrid, PA & NJ, United States

- Industry Thought Leadership: Selected twice as a featured speaker for the Apollo GraphQL Summit (2022 & 2024), presenting to audiences of 500+ attendees. Shared enterprise architectural strategies on scaling Apollo Federation, applying domain-driven design, and establishing RFC-driven schema standards, presenting both independently and collaboratively with engineering teammates.
- Established Enterprise API Governance: led and drove the adoption of GraphQL Federation standards across dozens of engineering teams, ensuring highly available, secure, and compliant architectural patterns within a complex financial services environment.
- Enhanced Developer Enablement (DevEx): Architected self-service implementation frameworks and practical guides, accelerating internal GraphQL adoption and reducing developer onboarding time by an estimated 12 months.
- Innovated API Infrastructure: Co-authored issued patent for a GraphQL Gateway-as-a-Service solution designed to streamline enterprise API management, decouple legacy systems, and modernize data routing at scale.
- Drove Firm-Wide Technology Adoption: Led enterprise-wide upskilling for about 160 engineers by founding an internal technology conference and establishing a formal GraphQL Certification Program to standardize competencies across distributed teams.
- Cultivated Agile Engineering Talent: Mentored engineers in architectural best practices and agile delivery, guiding direct reports from foundational tasks to autonomous, high-performance system design.

Principal Software Engineer

January 2020 - December 2021 (2 years)

Greater Boston

- Architected Conversational AI Integrations: Designed the integration layer for enterprise chatbot services, securely connecting external NLP models with internal infrastructure to deliver scalable conversational AI capabilities.
- Engineered Data Access Layers: Developed robust GraphQL polyglot resolvers to securely aggregate distributed customer data, creating a unified and performant access layer across disparate backend systems.
- Designed Multi-NLP Abstraction Patterns: Built a transparent multi-NLP integration framework, establishing a reusable architectural pattern that abstracted vendor complexity, streamlined the onboarding of new chatbots, and reduced operational overhead.

- Enhanced Platform Modularity: Designed and implemented decoupled micro-processes and services, significantly improving the core platform's modularity, extensibility, and fault tolerance.
- Optimized Cloud-Native Delivery: Drove the deployment and optimization of cloud-native applications, utilizing CI/CD pipelines (e.g., Concourse, Jenkins) to automate builds, ensure high availability, and maintain strict system reliability at scale.

Senior Software Engineer

December 2018 - January 2020 (1 year 2 months)

Boston, Massachusetts

- Engineered Cloud-Native Microservices: Designed and developed scalable, high-impact Node.js services within a distributed environment to securely orchestrate complex internal and external system calls for enterprise chatbot integrations.
- Accelerated Delivery Pipelines: Built and optimized CI/CD workflows utilizing Jenkins and Concourse to automate testing and deployment, significantly reducing delivery cycles while enforcing strict code quality standards.
- Drove Engineering Quality: Ensured rigorous platform reliability by implementing comprehensive testing strategies; spanning unit, functional, regression, and performance testing, to guarantee system maintainability at scale.
- Led End-to-End System Delivery: Directed full-lifecycle software development, combining agile methodologies with critical architectural research to consistently deploy resilient, distributed systems.

Software Developer

May 2015 - December 2018 (3 years 8 months)

Greater Boston

- Drove Technical Solution Design: Led requirements gathering and system architecture alignment between the Virtual Assistant team and the Chapter of Excellence, establishing robust integration patterns for enterprise chatbot services.
- Engineered Core Platform Integrations: Integrated the Virtual Assistant architecture into the core enterprise platform (using REST/APIs/Node.js), enabling secure, real-time data retrieval for account balances and dynamic, personalized user experiences.
- Directed Technical Delivery: Managed the end-to-end technical execution of high-impact customer features, bridging stakeholder requirements with agile engineering implementation to ensure transparent, on-schedule deployments.

Santander Bank with Everis

Software Developer

September 2011 - May 2015 (3 years 9 months)

Greater Boston

- Enterprise Application Maintenance: Developed and maintained enterprise applications across multiple teams, focusing on enhancing legacy systems and supporting ongoing operational needs.
- Backend Stability & Performance: Updated and maintained backend applications, ensuring stability and consistent performance in a production environment.
- Data Processing Solutions: Built standalone applications using a variety of technologies to handle database connectivity and data processing.
- Cross-Platform Legacy Support: Collaborated on diverse legacy projects, utilizing Microsoft and Java technologies to manage large-scale data processing and online application requirements.

Lara Web Works

Web Developer/Designer

September 2009 - September 2011 (2 years 1 month)

- Business Operations: Founded and operated a web design business, building sites for small businesses aiming to increase sales and online visibility.
- Web Design & Development: Provided technical services including website design, web application development, SEO, and digital marketing to help clients reach their target markets.
- Client & Project Management: Managed client relationships, project delivery timelines, and ongoing technical support from start to finish.

Education

University of Massachusetts Amherst

Master of Computer Applications - MCA, Computer Science · (February 2023 - September 2023)

Tiffin University

Bachelor's degree, Computer Science · (January 2020 - December 2021)

University of Massachusetts Lowell

Information Technology (AS), Information Technology · (September 2014 - May 2018)

Strayer University

Diploma, Web Development · (2005 - 2007)